Estimating truck fuel consumption with machine learning using telematics, topology and weather data

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***Abstract*— When purchasing new trucks it is crucial to have a good estimate of the future fuel consumption of the different truck models to choose from. The fuel consumption is a function of various external variables like the total weight, the average speed and its variation, topography, weather conditions and others. As opposed to classic approaches using physical models, the research presented in this paper is using machine learning techniques to predict fuel consumption based on independent variables. Most of the used data is provided by telematics systems from more than 500 trucks over two years. Three different ML models are compared and gradient boost proves to be the best method for the scope of the investigated truck categories. Besides obvious significance of features like total weight and topography, the results confirm that available data on weather conditions also help to improve the forecast. The obtained quality of the fuel prediction, measured as rooted mean squared error, outperforms results reported in prior research on this topic.**

***Keywords*—Fuel consumption, Machine Learning, Artificial Intelligence, Long haul transportation, Telematics, Weather**

I. Introduction

Reducing CO_2 emissions has become a major concern in road transportation. For combustion engines of a given type of fuel (diesel, gasoline or liquefied natural gas), those emissions are proportional to the consumed amount of fuel. Therefore, trucking companies have a twofold motivation to operate their vehicles in a way that minimizes fuel consumption: a monetary and an environmental one. There is a variety of factors that influence fuel consumption: travelling speed, acceleration, road gradient, weather, total weight, driving style of the driver, traffic conditions, configuration of the traction unit (motor + powertrain), road surface and even tyre models. Some factors, like the weather, are out of control to the managerial decision. Others, like the driver behavior, can be influenced at least indirectly [1].

The motivation for the research presented in this paper was to assess the impact of different truck models on mean fuel consumption given a specific use case, i.e. predefined trips and associated payload. Depending on the context, choosing an appropriate truck can be an operational or a strategic decision. It is operational when the truck is selected from an existing heterogeneous fleet and is affected to a single transportation order. But it can also be strategic when the purchase decision for the truck is taken with respect to a long term contract that will imply using this vehicle under the same conditions for a longer time. In both cases companies are interested in the average fuel consumption as a function of the chosen truck model.

The traditional approach to model fuel consumption as a function of independent variables consists of setting up physics based analytical models. Early work on this for cars is presented in [2] and subsequent authors established standard models also for heavy-duty trucks like STARS [3], MOVES [4] or VETO [5]. These models establish a physical relationship between the energy consumption and the forces acting against the traction unit. To fit specific types of vehicles they have to be parameterized using statistical models and data from a set of driving cycles. This limits their proper usage to the vehicle models for which they have been calibrated and also to the conditions of the driving cycles.

More recently [6] compared the simulated results by using the tool MIRAVEC which is based on VETO with real data collected by telematics systems of light, medium and heavy trucks on motorways in England. The results indicated a significant deviation between the simulated and the observed values. A comparison with a second model (HDM-4) yielded even bigger differences. The author concludes that the use of these models requests specific validation and calibration for both, the considered vehicles, as well as the prevailing conditions (climate, road), which is a time consuming task.

An alternative approach for estimating fuel consumption is provided by regression models that do not rely on physical relationships but on the ideas of machine learning (ML). Before being applied to road transportation, ML methods proved their usefulness in other domains [7]. [8] proposed ML-based regression models by combining smartphone data (GPS, speed, acceleration, road inclination) with data received from the OBD-interface (fuel consumption) for passenger cars. Out of a variety of tested regression models Random Forest and Boosting (cf. Section III) performed the best. As for heavy duty trucks, [9] conducts ML-based work in the framework of a project on the dynamic formation of truck platoons. Data is limited to trucks operated by Scania's Transport Laboratory in Sweden. Besides comprehensive information on the truck properties and the road network, the model also includes weather and driver behaviour data. Other authors also report on ML-based fuel consumption prediction, but the results were obtained under different settings or for other vehicle types, e.g. buses ([10], [11]). The main contribution of this paper in comparison to [9], [10] and [11] lies in the broader range of exploited data with respect to truck models and geographic coverage and in the significantlly higher accuracy of the obtained predictions for fuel consumption.

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The paper is organized as follows: Section II describes the used data sources and applied data transformation as well as data cleaning. Section III explains the applied ML techniques and is followed by a detailed description of the training and validation setting together with the presentation of the numerical results in Section IV. Section V summarizes the findings and gives an outlook to possible further research.

II. DATA SOURCES AND FEATURE PROCESSING

Vehicle data were obtained from European long-distance haulage companies using the Fleetboard telematics system implemented in heterogeneous fleets [12]. In total, 5.681.815 data points of 508 trucks over the years 2017 to 2018 were processed. Each data point contained a time stamp, the position, total fuel consumption since the vehicle has been put into service and total weight. Most often (79%), the time interval between two subsequent data points was 10 minutes. Intervals not matching 10 min were discarded in the further data analysis to avoid irregularities. Thus, two subsequent data points represented route segments ('tracks') for which, in a second step, a number of relevant features (predictor variables) were determined in order to estimate the fuel consumption (dependent variable) of this route segment.

The fuel consumption for the 10 min tracks was calculated as the difference between the total fuel consumption at the end point of the segment and its starting point. These total fuel consumption values were available with an accuracy of millilitres or in half liter steps, depending on the truck manufacturer.

To account for systematic deviations between the fuel consumption reported by the telematics system and the real fuel consumption, the billed amount of fuel for each vehicle was compared to its reported fuel consumption over a longer time span. This way, a vehicle specific correction factor could be determined that was applied to all its total fuel values first in order to improve data quality.

A feature with significant impact on fuel consumption is the total weight. It is directly reported by the telemetric system. The number and intensity of accelerations along a route segment is another important predictor. As it was not possible to get this information directly, the variance of speed was used as a substitute. The variance was calculated using so called 'delta positions' inside a route segment. Actually, in addition to the 10 min interval data which includes a variety of measures (like fuel consumption, weight etc.), the telematics system transmits GPS coordinates every minute. Using these delta positions, it was possible to compute the average speed values for each 1 min interval. These values were then used to calculate the variance of speed during the 10 min interval.

The fleet of vehicles consisted of medium to heavy trucks of three different manufacturers (Mercedes-Benz, MAN and Scania). Looking at the 508 trucks for which data were available, each truck can be assigned to one of 16 different *truck models*. A truck model shares at least the following properties: manufacturer, engine power, number of axles, cabin type and trailer type. The *truck model* defined this way is also used as a feature in the machine learning models that will be presented in the next section. These 16 truck models can be grouped into three different categories of trucks: 20t articulated trucks, 40t articulated trucks and 40t semi-trailer trucks (see Table I). The truck categories are only shown here to illustrate the type of trucks that were used in this research and have no importance for the modelling process.

TABLE I. CLASSIFICATION AND NUMBER OF THE TRUCKS

Truck category	Number of truck models	Number of individual trucks	Example of a truck model
20t articulated trucks	3	187	MAN TGL 8.220 4x2
40t articulated trucks	3	71	MB 2542 LNR
40t semi-trailer trucks	10	250	Scania R410 Topline

All positions were mapped onto the street network of the OpenStreetMap project [13] to gather the road types traveled (motorway / primary roads / other) and the ratio of the distance travelled in 10 min for which these roads were situated in urban areas. As a side benefit of this process, it was possible to calculate the distance driven based on the street network without relying on the less accurate mileage measurements of the vehicle.

Since OpenStreetMap data only contains sparse altitude indications, elevation data of the NASA SRTM 1 arc second global data set were merged with the street network [14]. Tunnels and bridges that do not follow the course of the earth's surface were detected and their altitude values levelled out in a later processing step.

Finally, weather data of the German Meteorological Service (DWD) was used to gain information about ambient temperature, air pressure, wind and precipitation [15]. Hourly measurements of about 500 weather stations were used to interpolate meteorological conditions of the analysed route segments. Because these weather stations were located in Germany only, the vehicle positions analysed were also constrained to German national borders. Missing weather data was filled by approximation to the nearest available weather data using a simple K-nearest-neighbours regression on time and space distance (cf. Section III.). A complete list of the used features is shown in Table II.

TABLE II. LIST OF THE USED FEATURES

Category	Features (per 10 min track)
main features	total weight [t], truck model [categoric], average speed [km/h], engine power [HP], height meters (increasing, decreasing, difference) [m]
road features	share of motorway [%], share of primary road [%], share of secondary and tertiary streets [%], share of roads within built-up areas [%], longitude [°], latitude [°]
acceleration	variance of the speed (inferred from delta positions between the regular positions items in 1 minute intervals) [m^2/s^2]
weather features	temperature [°C], wind speed [m/s], relative wind direction [°], precipitation [L/m^2], air humidity [%], air resistance [N]

During the data cleaning process, a minor part of the data had to be removed due to feature inconsistencies, e.g. unrealistic average speeds. After all, 3.8 million data records out of the initially more than 5 million were kept. The following plots illustrate the distributions of the total weight, the average speed, and of the fuel consumption after the elimination of outliers.

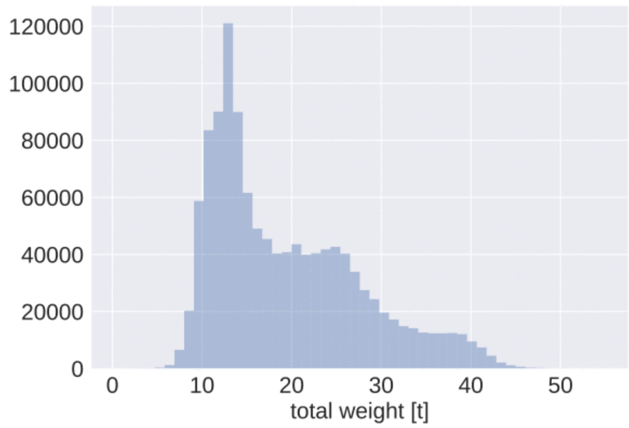

Fig. 1. Distribution of total weight

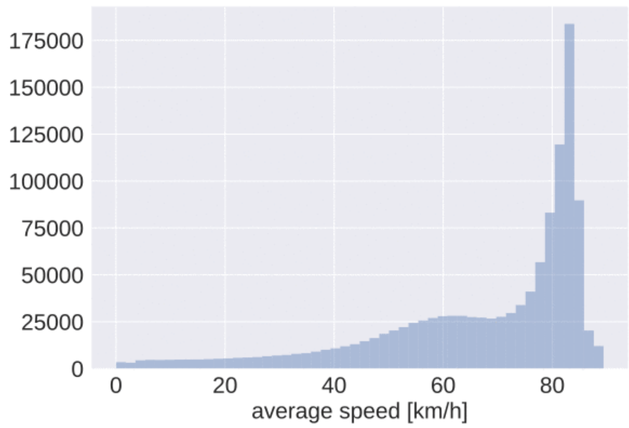

Fig. 2. Distribution of average speed

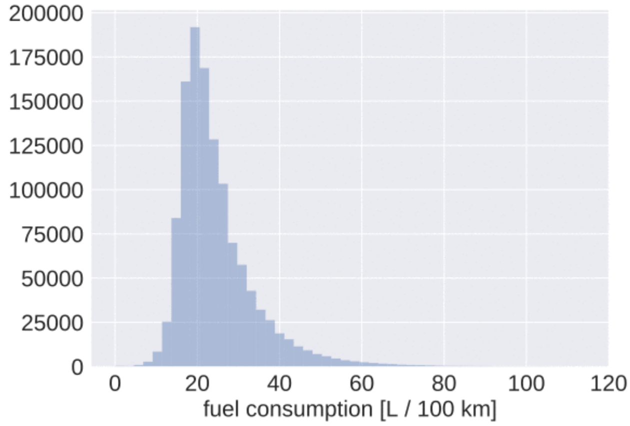

Fig. 3. Distribution of consumed fuel

The different numerical variables were then normalized by quantile normalization [16]. As a consequence, less influence by extreme values was expected. As target distributions of the normalization a normal distribution or a uniform distribution were used, depending on the distribution of the feature to be normalized. Furthermore, as the vehicle model is a categorical feature we had to transform it into a binary representation using one hot encoding.

III. MACHINE LEARNING MODELS

For predicting the fuel consumption several regression models can be taken into account. Regression models in general are tools to predict numerical values. Given the normalized values of the features of the 10 min-sections, as well as the matching true values of the fuel consumption, the regression model is trained. After that, the model is simply a set of calculation rules to predict the fuel consumption given any feature values.

In this section of the paper we shortly describe how the three chosen regression models work, i.e. how the training is done for each model. However, the models were not implemented from scratch, instead standard implementations of the models from Python libraries were used. As a description of each model type would be beyond the scope of this contribution, references to specialized papers are given for each model type.

Prior research [9] has shown that boosting methods are a suitable method for predicting the fuel consumption. Other widely used regression models are (deep) feed forward neural networks. Therefore, both methods were used in our comparison. To draw from a wider range of model families and since it is very likely that two trucks of the similar type, with similar speed, weight, and so on have a very similar fuel consumption, we chose the instance based k-nearest-neighbors regression model that searches the most similar trucks in the data for the predictions. The next paragraphs will briefly describe these three models and the hyperparameters of each model to be optimized and will be followed by a comparison of their performances in Section IV.

K-nearest-neighbors regression [17]

Given the training set and the training labels described in Section II., k-nearest-neighbors estimates the fuel consumption of an unknown entry in the test set by finding the K entries in the training set having the smallest distance to the unknown entry. The estimated value is then the (weighted) average of the labels of these K entries.

The main drawback of this method is that one has to adjust the importances of the features during the tuning of the hyperparameters. All features have to be multiplied with a certain factor to optimize the predictions for the test data. E.g. since the impact of the total weight on fuel consumption is certainly higher than the impact of the ambient temperature, the total weight must be multiplied with a higher factor than the temperature. The only way to find these factors is optimising them as hyperparameters with the help of a validation set. The other tuned hyperparameters are K, the used metric and the decision, whether to use weighted or unweighted averages for the prediction.

Feed forward neural network regression (ANN) [18]

Neural networks are inspired by the human brain, meaning that they consist of single elements (called neurons) that are placed layerwise (one input layer, >= 1 hidden layers, and one output layer of one neuron) in a network. Each neuron of a hidden or output layer computes its own output by feeding the weighted outputs coming from all neurons in the previous layer to a so called activation function. The input layer contains as many neurons as there are features, each of them has one feature value that it sends to the next layer. Given the weights in the network and a sample of the data, the neural network computes one output value. If the weights were optimal, the output would correspond to the observed value.

In order to use such a network for the solution of regression problems, all the weights of the network have to be optimised. This is done by presenting samples from the training set to the network and compute the actual output of the network. The weights can be optimized by back-propagating through the network and adjusting the weights regarding the difference between the computed output and the actual values. This is repeated until the network has seen all the training data n times, n being a hyperparameter. Other hyperparameters are the number of the hidden layers, the number of neurons per hidden layer, the activation functions and the dropout rate, a parameter that reduces overfitting.

Gradient boost regression [19]

Function approximation by gradient boosting performs a weak estimation for a certain number of iterations such that the sum of all obtained estimates is then the overall approximation of the function. The full algorithm starts with an initial estimate of the objective function, e.g. the mean of all training labels. Then in each step the negative gradient of the loss function regarding the actual estimate (cf equation (13) in [19]) is computed. This gradient is predicted by a weak model. Usually a simple decision tree model is chosen. The prediction is multiplied with a certain learning rate and added to the previous estimate to get the new actual estimate.

The main decisions to be taken when creating a gradient boost model are the choices of the weak model and of the loss function. In this study the most popular model, namely decision trees, was chosen. The mean squared error was used for the objective function. Hyperparameters of a gradient boost model are the parameters of the underlying decision tree model, such as e.g. the maximum tree depth and the fraction of the features that is used for each tree. Furthermore, the learning rate and the number of estimation iterations can be tuned.

IV. Tests and Numerical Results

To explore the performance of the three machine learning approaches, the data was split in a training set and a test set by randomly choosing 31% of the vehicles as the test set. The data from the other 69% of the vehicles were exclusively used for training. These percentages are the result of a stratified sampling from the different truck models. For each model we sampled a third (rounded down) of the vehicles of this model to form the test set. E.g. for a truck model X there existed entries of 20 different vehicles. Then six vehicles were randomly chosen and all entries of these vehicles were assigned to the test set, while all remaining entries formed the training set. As a consequence, testing was done with data from trucks that were not subject of the training phase. However, we did not test our ML-models for new vehicle models and all truck models were at least represented by nine trucks.

For the comparison of the performance of the different regression models and feature categories we initially used only the data of 2017 (1.6 million data records). This was due to the fact, that during the first phase of the study only those data were available. In the experiments the data of the selected features were fed into one of the regression models and the hyperparameters of the model were tuned. To do so, we performed a 10-fold cross validation and optimized the hyperparameters by a random grid search. For the objective function we used the mean squared error.

We did several experiments to assess the performance of the models. First, the different techniques from Section III were compared. The results obtained by each of the models were evaluated using the mean absolute error (MAE), the rooted mean squared error (RMSE) and the coefficient of determination R^2. MAE and RMSE refer to the fuel consumption in l / 100 km.

According to Table III the gradient boost model shows the best performance compared to the K-Nearest-Neighbours regression and the Artificial Neural Net regression. However, the performance of the three models was rather close. Nonetheless, for the subsequent experiments on the importance of the different feature groups, we concentrated on the gradient boost model as it performed best.

TABLE III. Comparison of three ML techniques on 2017 data

Model	MAE	RMSE	R^2
K-nearest neighbors	3.18	5.10	0.724
Neural nets	3.11	4.80	0.758
Gradient boost	2.97	4.63	0.771

Furthermore, an analysis of the computational times, more precisely the training time and the prediction time of these three models was performed. The training time comprises the time that is needed to train the model, it does not include the time that is needed to optimize the hyperparameters. The prediction time represents the time that is needed for one prediction of the fuel consumption given the normalised feature values. For practical applications the prediction time is most critical, as opposed to the the training time which is only done once upfront. According to Table IV the prediction time is below 1 ms for all the three models with the Neural Network model beeing the fastest.

TABLE IV. Computational Times

Model	Training time [s]	Prediction time[ms]
K-nearest neighbors	2605	0.443
Neural nets	1610	0.0324
Gradient boost	365	0.12

A second series of experiments showed the importances of the different feature categories (Table V). We started with a gradient boost model using just the 'main features' (cf. Table II). We optimised the hyperparameters of the model and predicted the fuel consumption for the entries in the test set. Then we added the 'local features' and repeated the procedure. In a third step we added the 'variance of speed' and at last we added the 'weather features'. We computed the same metrics as before. The last row in Table IV shows the results when using the complete data set (2017 + 2018) with all available features.

TABLE V. INCREMENTAL IMPACT OF FEATURE GROUPS

Feature category	Data scope	MAE	RMSE	R^2
main	2017 only	2.97	4.63	0.771
+road	2017 only	2.70	4.35	0.798
+variance of speed	2017 only	2.58	4.14	0.817
+weather (All features)	2017 only	2.48	4.05	0.825
All features	2017 and 2018	2.32	3.95	0.838

As one can see in Table IV, all feature categories have a considerable influence on the prediction quality. The best prediction for the chosen test set has a rooted mean squared error of 3.95. The quality of the predictions for this case is visualized in Fig. 4. Comparing the results to prior research, a significant improvement can be found. For time intervals of 10 minutes [9] found a RMSE of 7.4 which is almost twice as high as the corresponding value in the last row of Table IV.

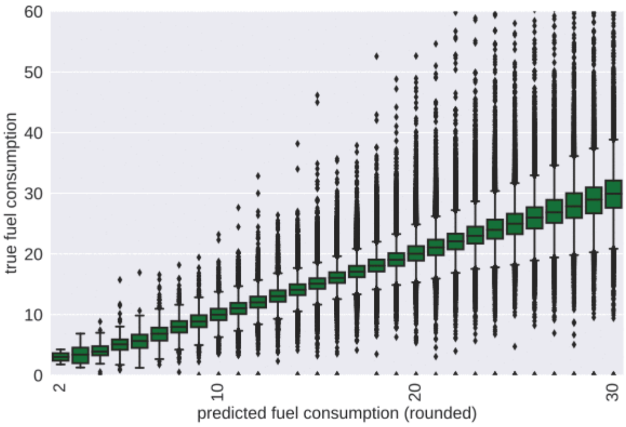

Fig. 4. Regression function predicted vs. true fuel consumption (l/100km)

Furthermore, [10] predicted the fuel consumption with data from trucks with constant speed of 80 +/- 2.5 km/h using time intervals of one minute. To have a fair comparison with our results, we report the RMSE of our results as a function of the average speed. Fig. 5 shows the quality of the predictions for selected average speeds. As expected, the predictions are best for high average speeds, usually in combination with little speed variations. For an average speed of 80 km/h Fig. 5 shows a rooted mean squared error of 2.16 in comparison to 4.64 in [10].

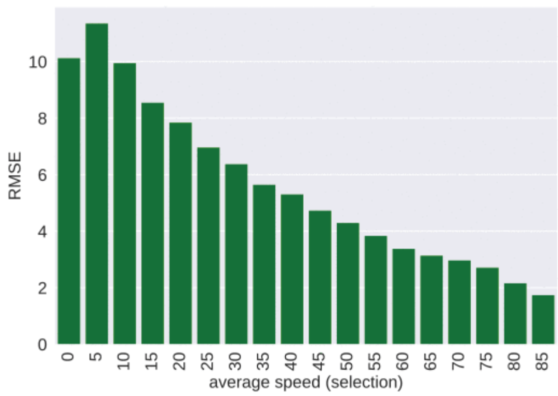

Fig. 5. Rooted mean squard error as a function of mean speed

V. CONCLUSION

In this study machine learning techniques have been applied to predict fuel consumption of heavy duty trucks. Two years of data of over 500 single trucks representing 16 different truck models were available. A comparison of three different ML techniques indicated that the gradient boost model outperformed ANN and k-nearest-neighbors. Furthermore, the experiments provided evidence that besides the truck model, average speed, speed variance, road type and total weight, weather features also contribute to a significant improvement of the forecast. Compared to results obtained in similar studies, the developed model provided RMSE values that were considerably lower than those reported in prior research.

Directions for further research could be to include additional features. Due to private data protection, it was not possible to get information about the assessment of the driver behaviour for this study, which presumably has also an impact on fuel consumption. Another interesting feature is tyre pressure, as this has an impact on the rolling resistance of the vehicle. Although pressure sensor systems exist, the trucks used in this study were not equipped with such technology. The same is true for road surface characteristics.

In the upcoming age of electrified trucks with still limited battery autonomy, it would be interesting to provide reliable tools to estimate the remaining kilometers a truck can travel given the exact route to its next destinations. The approach applied in this research can also be adapted to this kind of vehicles.

ACKNOWLEDGEMENTS

The presented research was supported under the mFund program of the German Federal Ministry of Traffic and digital Infrastructure.

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